COMPILING THE ALMGREN–PITTS COMBINATORIAL ARGUMENT

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Abstract. The Almgren–Pitts combinatorial argument — the cover-refinement plus bookkeeping backbone of Step 2 in min-max constructions of minimal hypersurfaces — is a recurring technical step that each author re-derives in local notation. We extract it from the literature as a standalone metric-combinatorial algorithm and formalize it in Lean 4 as the importable library `CombArg`.

1. Introduction

A minimal hypersurface is a critical point of the area functional in codimension one. On a closed Riemannian manifold (M^{n+1}, g), direct minimization over a fixed homology class may produce only degenerate configurations, and the relevant critical points are typically of saddle type. The resolution, following Birkhoff (1917) for closed geodesics and Almgren–Pitts for higher-dimensional varifolds, is a *min-max* construction; regularity of the output is Schoen–Simon [SS81].

Theorem 1.1: Almgren–Pitts min-max theorem [Pit81, SS81]

Every closed Riemannian manifold (M^{n+1}, g) admits a nontrivial embedded minimal hypersurface $\Sigma \subset M$ whose singular set $\operatorname{Sing}\Sigma$ has Hausdorff dimension at most $n-7$.

The core technical result of Pitts's proof — used in every subsequent treatment [CDL03, DLT13, DLR18, DL16, MN14] — is the Almgren–Pitts combinatorial argument, stated in the Colding–De Lellis formulation as follows.

Proposition 1.2: Almgren–Pitts combinatorial Argument, [CDL03, Prop. 3.4]

Let Λ be a homotopically closed family of sweepouts of M^{n+1}, and $m_0 = \inf_\Lambda \mathcal{F}$. There exists a min-max sequence $\{\Gamma^N\}$ converging as varifolds to a stationary integral varifold of mass m_0, with Γ^N $1/N$-almost-minimizing in every pair $(U^1, U^2) \in \mathcal{CO}$ of $\mathcal{H}^n$-disjoint open sets, for all N sufficiently large.

The proof of Proposition 1.2 reduces to a one-parameter claim on the sweepout [DLT13, §3.2]:

> **Claim.** For each N sufficiently large, there exists $t_N \in [0,1]$ such that $\Gamma^N := \Gamma^N_{t_N}$ is $1/N$-almost-minimizing in every $(U^1, U^2) \in \mathcal{CO}$ and $\mathcal{H}^n(\Gamma^N) \geq m_0 - 1/N$.

The Claim is proved by contradiction: from its negation, every t with $\mathcal{H}^n(\partial\Omega^N_t) \geq m_0 - 1/N$ admits a *local replacement* with saving $1/(4N)$ [DLT13, Lemma 3.1]; a combinatorial cover-refinement and bookkeeping argument on $[0,1]$ ([DLT13, §3.2] Steps 1–2) then assembles from these local replacements a modified sweepout with supremum strictly below m_0, contradicting $m_0 = \inf_\Lambda \mathcal{F}$. The combinatorial argument depends on no geometric measure theory, no admissible class, and no ambient regularity; it is a bookkeeping construction on continuous real-valued functions, yet has no canonical reference.

Stripped of its GMT instantiation, the content of Steps 1–2 is the following scalar statement, which is the precise theorem we formalize.

Date: April 23, 2026.
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Theorem 1.3: Extracted scalar version of Steps 1–2

Let X be a pseudo-metric space with an abstract paired-cover structure. Let $f : [0,1] \to \mathbb{R}$ be continuous with $m_0 = \sup f$ and $m_0 > 0$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose that at every $t \in [0,1]$ with $f(t) \geq m_0 - 1/N$ there is a *local witness*: an open neighborhood $U_t \ni t$, a paired cover c_t in X, and a continuous replacement energy $E_t : [0,1] \to \mathbb{R}$ satisfying $f(s) - E_t(s) \geq 1/(4N)$ for every $s \in U_t$. Then there exists $f' : [0,1] \to \mathbb{R}$ with

$$\sup f' \;\leq\; m_0 - \frac{1}{4N}.$$

The boundary $\partial\Omega_t$ is replaced by a continuous $f : [0,1] \to \mathbb{R}$; the Hausdorff-measure hypothesis becomes $m_0 = \sup f$; the nested pair (A_t, B_t, U'_t) collapses to a single neighborhood U_t; the replacement family $\{\partial\Omega_{t,\tau}\}$ collapses to a single continuous E_t; the almost-minimizing hypothesis disappears. What remains is the combinatorial bookkeeping. Theorem 1.3 reappears in §2.3 as the library's public one-parameter API (Theorem 2.8); it is a specialization of an abstract core (Theorem 2.7) parametric in the sweepout space and in a pair (δ, ε) replacing the hardcoded $(1/N, 1/(4N))$.

Scope. This note formalizes Steps 1–2 of the proof of the Claim as a Lean 4 library, `CombArg`. We do *not* formalize [DLT13, Lemma 3.1] (the replacement family), the $(U^1, U^2) \in \mathcal{CO}$ disjoint-pair structure, or the final contradiction against $m_0 = \inf_\Lambda \mathcal{F}$; these depend on the GMT and admissible-class framework of the ambient construction and remain the consumer's responsibility. The library ships a single abstract theorem (§2.2) parametric in the sweepout space together with a one-parameter specialization to $[0,1]$ (§2.3); the formalization is machine-verified against `Mathlib`, declares no custom axioms, and contains no `sorry`. Source: https://github.com/MathNetwork/comb_arg.

2. Specification

We present the library specification following standard software documentation conventions: interface first (§2.1), public API (§2.2–§2.3), internal lemmas (§2.4).

2.1. Interface: types and structures.

Throughout, `K` denotes a topological space (the parameter space), `X` a pseudo-metric space (the ambient space whose covers witness replacements), and `f : K → ℝ` a continuous scalar energy.

Definition 2.1: `PairableCover` class

`CombArg.PairableCover` A `PairableCover X` instance declares, for each abstract `c : Cover`, a left region and a right region in X and a scalar diameter, subject to two axioms: the two regions are disjoint, and overlapping combined regions of diameter-comparable covers are ordered by inclusion.

Definition 2.2: `LocalWitness` structure

`CombArg.LocalWitness` A `LocalWitness K X f t` ε carries an open neighborhood $U \ni t$ in K, a cover `c : Cover`, a continuous replacement energy $E : K \to \mathbb{R}$, and the inequality $\forall s \in U,\ f(s) - E(s) \geq \varepsilon$.

Definition 2.3: Near-critical set

`CombArg.Refinement.nearCritical` For continuous $f : K \to \mathbb{R}$, scalar m_0, and $\delta \geq 0$, the δ-near-critical set is $\{t \in K : f(t) \geq m_0 - \delta\}$. When $\delta = 1/N$ we write NC.

Definition 2.4: `FiniteCoverWithWitnesses` structure

`CombArg.FiniteCoverWithWitnesses` A `FiniteCoverWithWitnesses K f m₀ δ ε` carries a finite nonempty index type ι, pieces $\{p_l \subseteq K\}_{l\in\iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \in \mathbb{R}_{>0}\}$, subject to four conditions: (I) $f - E_l \geq s_l$ on p_l; (II) each $t \in K$ lies in at most two pieces; (III) $s_l \geq \varepsilon$ for all l; (IV) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$. This is the abstract form of the refined covering $\{\overline{J_l}\}$ produced at the end of [DLT13, §3.2 Step 1]; conditions (II)–(III) are DLT's two-fold overlap and per-piece saving invariants.

Definition 2.5: `InitialCover` structure

`CombArg.Refinement.InitialCover` Specialized to $K = \mathbf{I}$. An `InitialCover X f m₀ N` carries a finite family of open intervals $I_i = (c_i - r_i, c_i + r_i) \cap \mathbf{I}$ for $i = 1, \ldots, n$, each paired with a witness center $w_i \in \mathrm{NC}$ and a local witness at w_i with saving $1/(4N)$, subject to (a) spacing: $c_i + r_i < c_{i+2} - r_{i+2}$ for all valid i; (b) containment: $I_i \subseteq U_{w_i}$; (c) coverage: $\mathrm{NC} \subseteq \bigcup_i I_i$.[a] Corresponds to the initial covering $\{I_i\}$ of NC entering [DLT13, §3.2] *before* the Step 1 refinement.

[a] Conditions (a) and (b) transcribe [DLT13, §3.2]'s spacing and subset conditions; the pair (c_i, w_i) separates interval centers from witness centers, required because grid points need not themselves be near-critical.

Definition 2.6: `PartialRefinement` structure

`CombArg.Refinement.PartialRefinement` Given `ic : InitialCover` and $\ell \in \mathbb{N}$, a `PartialRefinement ic ℓ` carries pieces $J_1, \ldots, J_\ell \subseteq \mathbf{I}$ and an index assignment $\sigma : \{1, \ldots, \ell\} \to \{1, \ldots, n\}$ with (i) $J_k \subseteq I_{\sigma(k)}$ for each k; (ii) $I_{\sigma(k)} \subseteq \bigcup_{k'} J_{k'}$ for each k. This captures the *mid-induction state* of [DLT13, §3.2 Step 1], where each I_j has been either kept whole ("the same") or split ("into two halves").

2.2. Public API: abstract core.

Theorem 2.7: Abstract core

`CombArg.exists_sup_reduction_of_cover` **Input.** A compact, nonempty topological space K; a continuous $f : K \to \mathbb{R}$; $m_0 = \sup f$; scalars $0 < \varepsilon \leq \delta$; a `FiniteCoverWithWitnesses`$(K, f, m_0, \delta, \varepsilon)$ (Definition 2.4).
Output. There exists $f' : K \to \mathbb{R}$ with $\sup f' \leq m_0 - \varepsilon$.

2.3. Public API: one-parameter application.

Theorem 2.8: One-parameter application

`CombArg.exists_sup_reduction` **Input.** A pseudo-metric space X with a `PairableCover X` instance (Definition 2.1); a continuous $f : \mathbf{I} \to \mathbb{R}$; $m_0 \in \mathbb{R}$ with $m_0 > 0$ and $m_0 = \sup f$; $N \in \mathbb{N}$ with $N > 0$; for every $t \in \mathrm{NC}$ a `LocalWitness` at t with saving $1/(4N)$ (Definition 2.2).
Output. There exists $f' : \mathbf{I} \to \mathbb{R}$ with $\sup f' \leq m_0 - 1/(4N)$.
Specializes Theorem 2.7 to $K = \mathbf{I}$, $\delta = 1/N$, $\varepsilon = 1/(4N)$.

2.4. **Internal lemmas.** The declarations below are consumed by Theorem 2.8; proofs are sketched in §3.

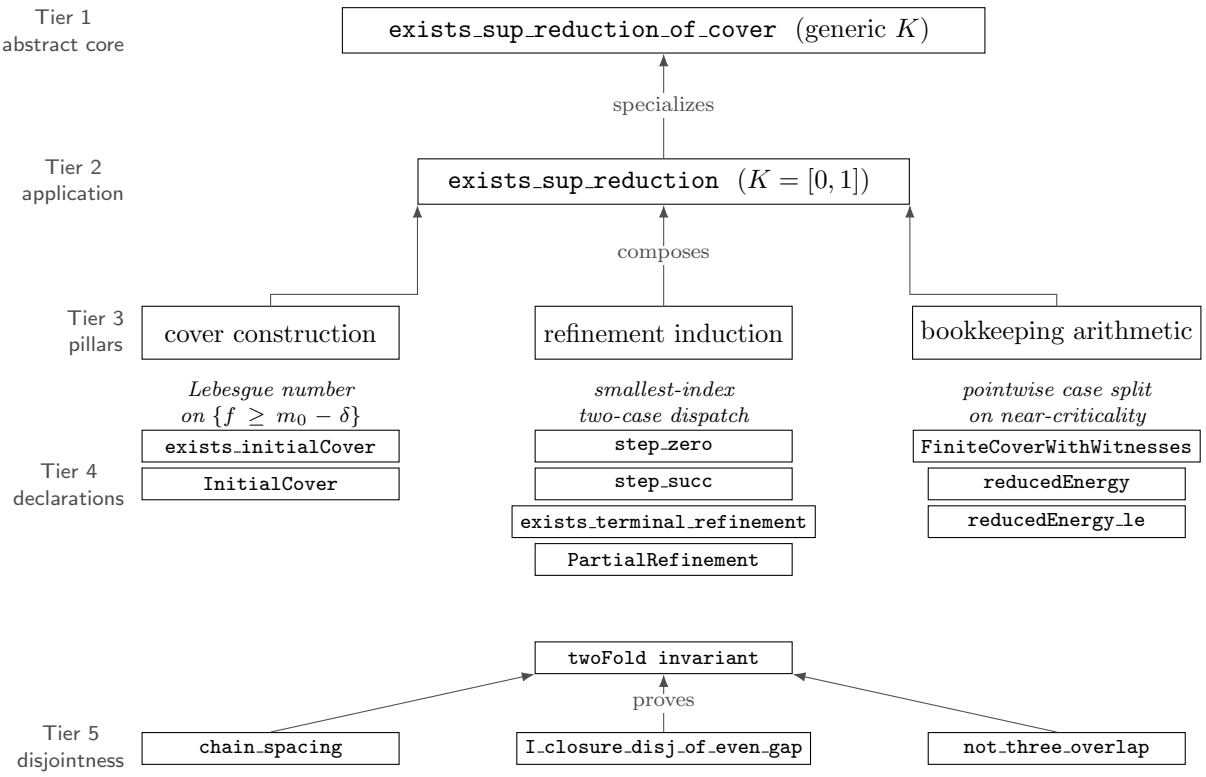

FIGURE 1. `CombArg` v0.1 architecture. The abstract core (Tier 1) is parametric in the sweepout space; the one-parameter application (Tier 2) instantiates $K = [0, 1]$ and composes three construction pillars (Tier 3). Each pillar's key declarations appear in Tier 4. The disjointness sub-module (Tier 5) supplies the two-fold overlap invariant consumed by bookkeeping.

Proposition 2.9: Compactness of NC

`CombArg.Refinement.isCompact_nearCritical`
`CombArg.Refinement.nearCritical_nonempty` If K is compact and f continuous, NC is compact. If additionally $m_0 = \sup f$ and $N > 0$, NC is nonempty.

- `CombArg.Refinement.exists_initialCover` — existence of an `InitialCover` via a grid plus Lebesgue-number construction on the compact sublevel set NC.
- `CombArg.Refinement.exists_terminal_refinement` — existence of a terminal `PartialRefinement` with injective σ via bounded smallest-index induction with two-case dispatch.
- `CombArg.Refinement.InitialCover.not_three_overlap` — at most two among any three distinct `InitialCover` intervals overlap at a common point (parity rescue).
- `CombArg.Refinement.saving_bound_closure` — the open-neighborhood saving bound in `LocalWitness` extends to the closure via continuity of E.
- `CombArg.Refinement.exists_refinement` — packages the 1D construction output as a `FiniteCoverWithWitnesses` with $(\delta, \varepsilon) = (1/N, 1/(4N))$ on $\mathbf{I}$; the input to Theorem 2.7.

Figure 1 shows the layered architecture.

3. Proof structure

We split the proof of Theorem 2.8 into three stages, each one mirroring a segment of [DLT13, §3.2]:

our stage	DLT13 §3.2 segment	output
Stage A	(setup, pre-Step 1)	`InitialCover`
Stage B	Step 1 (refinement of the covering)	terminal `PartialRefinement`
Stage C	Step 2 (conclusion: bookkeeping)	`FiniteCoverWithWitnesses`

Theorem 2.7 then consumes the output of Stage C and closes the chain.

3.1. Stage A: existence of the initial cover.

Proposition 3.1: Existence of an initial cover

`CombArg.Refinement.exists_initialCover` Under the hypotheses of Theorem 2.8, an initial cover (Definition 2.5) exists.

Construction. Compactness of NC (Proposition 2.9) plus the open cover $\{U_t\}_{t\in\mathrm{NC}}$ afforded by the witness hypothesis admit a Lebesgue number $\lambda > 0$: every λ-ball in NC fits inside some U_t. Choose $M \in \mathbb{N}$ with $M > 4/\lambda$, form the grid $c_k = k/M$ for $k = 0, \dots, M$, and select those c_k within $\lambda/4$ of NC. For each selected c_k pick a witness center $w_k \in \mathrm{NC}$ within $\lambda/4$ of c_k; the ball of radius λ around w_k fits inside some U_{w_k}, hence so does the interval $I_k := (c_k - 2/(3M), c_k + 2/(3M)) \cap \mathbf{I}$. Spacing condition (a) holds because consecutive selected grid indices differ by at least 2 in the selected finset, so their centers differ by at least $2/M > 4/(3M) = 2 \cdot (2/(3M))$. Coverage condition (c) holds because every $t \in \mathrm{NC}$ is within $1/(2M) < 2/(3M)$ of its rounded grid neighbor, which is selected. Condition (b) holds by the Lebesgue argument. □

3.2. Stage B: the refinement induction.

This stage realizes [DLT13, §3.2 Step 1]: each initial interval I_j with $j \geq 2$ is either kept whole ("remains the same") or split ("into two halves"), depending on whether the paired cover at $t_{\sigma(\ell)}$ is compatible with the new t_{i^*}. Figure 2 reproduces the [DLT13, Fig. 1] schematic.

Formally, $\{J_l\}$ is built by induction from step_zero (base case) via step_succ_at (inductive step).

Definition 3.2: Base case

`CombArg.Refinement.step_zero` step_zero(ic) is the length-1 partial refinement with $J_1 := I_1$ and $\sigma(1) := 1$.

Definition 3.3: Inductive step

`CombArg.Refinement.step_succ_at` Given a partial refinement pr of length $\ell + 1$ and an index $i^* \in \{1, \dots, n\}$ with $I_{i^*} \not\subseteq \bigcup_k J_k$, define a length-$(\ell + 2)$ extension by case split on whether $I_{i^*} \subseteq J_\ell \cup I_{\sigma(\ell)}$:

(Case 1) If yes, set $J_{\ell+1} := I_{i^*}$.

(Case 2) Otherwise, set $J_{\ell+1} := I_{i^*} \setminus I_{\sigma(\ell)}$.

In both cases set $\sigma(\ell + 1) := i^*$. Both invariants of Definition 2.6 are preserved; in Case 2 the preservation of (ii) uses the induction hypothesis on $I_{\sigma(\ell)} \subseteq \bigcup_k J_k$.[a]

[a] A variant `step_succ` chooses i^* internally via Finset.min′; `step_succ_at` takes i^* as an explicit argument and returns the extension bundled with the "i^* is covered" property, for termination.

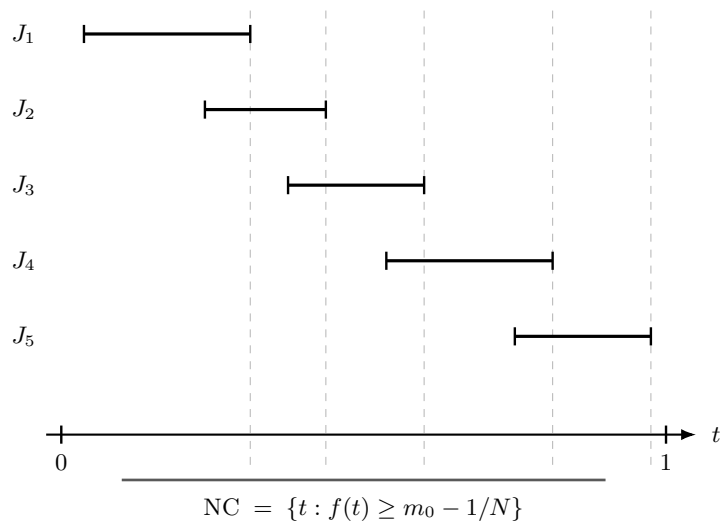

FIGURE 2. Staircase layout of the refined cover $\{J_l\}$ on $[0,1]$, after [DLT13, Fig. 1 left]. Each piece J_l is drawn on its own row so that vertical sweeps are legible: at any parameter $t \in [0,1]$, the number of rows the vertical line crosses is the overlap multiplicity; the cover satisfies $|\{l : t \in \overline{J_l}\}| \leq 2$ for every t (two-fold invariant). The NC bracket on the parameter axis marks the $1/N$-near-critical set, which is contained in $\bigcup_l J_l$.

Proposition 3.4: Termination

`CombArg.Refinement.exists_terminal_refinement` Starting from step_zero(ic) and iterating step_succ_at, one reaches in at most n steps a partial refinement of some length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i.

Argument. Let rem(pr) be the finite set of indices i with $I_i \not\subseteq \bigcup_k J_k$. One step of step_succ_at at $i^* \in$ rem(pr) (chosen as the smallest such) produces pr$'$ with $i^* \notin$ rem(pr$'$) and rem(pr$'$) $\subseteq$ rem(pr) $\setminus \{i^*\}$, hence strict decrease. Injectivity of σ follows from the case split in Definition 3.3: the new $\sigma(\ell+1) = i^*$ is fresh because $i^* \in$ rem while existing $\sigma(k)$ correspond to already-covered indices. □

3.3. **Stage C: assembly.** This stage realizes [DLT13, §3.2 Step 2]: from the refined $\{J_l\}$ together with the local-witness saving bound one packages a `FiniteCoverWithWitnesses` (Definition 2.4), whose three invariants (per-piece saving floor, two-fold overlap, coverage of NC) are precisely the data DLT's Step 2 bookkeeping consumes. Here we extend the saving bound from open J_k to $\overline{J_k}$ and derive the two-fold invariant from spacing (a). Theorem 2.7 then turns the assembled cover into the sup-reducing competitor f' (our formalization stops here; DLT proceeds to derive a contradiction against $m_0 = \inf_\Lambda \mathcal{F}$, which is the caller's responsibility).

Proposition 3.5: TwoFold via parity

`CombArg.Refinement.terminal_twoFold` For a partial refinement pr of length L with σ injective, every $t \in \mathbf{I}$ lies in at most two of $\overline{J_1}, \ldots, \overline{J_L}$.

Argument. From spacing condition (a), iterated m times, one obtains that indices i, j with $|i-j| = 2(m+1)$ (an even gap ≥ 2) have strict separation between $\overline{I_i}$ and $\overline{I_j}$, hence $\overline{I_i} \cap \overline{I_j} = \emptyset$ (`CombArg.Refinement.InitialCover.chain_spacing`, `CombArg.Refinement.InitialCover.closure_disjoint_of_even_g`

Suppose three distinct indices $k_1, k_2, k_3 \in \{1, \dots, L\}$ admit $t \in \overline{J_{k_1}} \cap \overline{J_{k_2}} \cap \overline{J_{k_3}}$. Since $\overline{J_k} \subseteq \overline{I_{\sigma(k)}}$ and σ is injective, $\sigma(k_1), \sigma(k_2), \sigma(k_3)$ are three distinct indices in $\{1, \dots, n\}$ with a common point in the closures of $I_{\sigma(k_i)}$. Among any three distinct integers $a < b < c$ some pair has even gap ≥ 2: either $c - a$ is even (and necessarily ≥ 2), or $c - a$ is odd and ≥ 3, in which case $b - a$ and $c - b$ have opposite parities and one of them is the even gap. The pair with even gap has disjoint closures, contradicting the common point (`CombArg.Refinement.InitialCover.not_three_overlap`). □

Proposition 3.6: Closure extension of saving bound

`CombArg.Refinement.saving_bound_closure` Let pr be a partial refinement. For every k and every $t \in \overline{J_k}$, $f(t) - E_{\sigma(k)}(t) \geq 1/(4N)$, where $E_{\sigma(k)}$ is the replacement energy of the $\sigma(k)$-th local witness.

Argument. On $J_k \subseteq U_{w_{\sigma(k)}}$ the witness gives $f - E_{\sigma(k)} \geq 1/(4N)$. The function $g := f - E_{\sigma(k)}$ is continuous (Definitions 2.2 requires E_t continuous; f is continuous by hypothesis). A closure point is a limit of a sequence in J_k, and g preserves $\geq$-inequalities under limits. □

Lemma 3.7: Assembly

`CombArg.Refinement.exists_refinement` A partial refinement pr of length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i yields a refinement of $(\mathbf{I}, X, f, m_0, N)$ in the sense of Definition 2.4, with $p_k := \overline{J_k}$, uniform saving $s_k := 1/(4N)$, and $E_k :=$ the replacement energy of the $\sigma(k)$-th witness.

Argument. Invariant (I) is Proposition 3.6. Invariant (II) is Proposition 3.5. Invariant (III) holds by construction ($s_k = 1/(4N)$). Coverage of NC chains $\mathrm{NC} \subseteq \bigcup_i I_i \subseteq \bigcup_k J_k \subseteq \bigcup_k \overline{J_k}$. □

Theorem 2.8 now follows: chain Propositions 3.1 and 3.4, invoke Lemma 3.7, and apply Theorem 2.7.

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